

Insert Generator — Quick Start

The one-window tool: type the genes, click a button, get the annotated insert

tcr_insert_gui.py does the entire pipeline in one step. You type the six TCR values into a window, click **Generate Insert**, and it runs Stitchr, trims the chains correctly (alpha to remainder 0, beta keeping the terminal G), builds the lentiviral insert, and opens a colour-coded annotated document in your browser. No manual trimming, no spreadsheets, no command typing per chain.

One-time setup

Make sure these are installed (you only do this once):

```
pip install stitchr
stitchrdl -s HUMAN
```

If you're on a Homebrew Mac and the app says Tkinter is missing, also run:

```
brew install python-tk
```

Running the tool

Step 1 — Open Terminal and point it at the folder

Type `cd` and a space, drag the bundle folder onto the Terminal window, and press Enter:

```
cd ~/Desktop/insert_bundle
```

Step 2 — Launch the app

```
python3 tcr_insert_gui.py
```

A window opens with fields for the alpha and beta chains.

Step 3 — Fill in the fields

Enter the sample name, a system label (e.g. "DR4 / clone 461"), and the six TCR values:

- **Alpha:** *TRAV gene, TRAJ gene, TRA CDR3*
- **Beta:** *TRBV gene, TRBJ gene, TRB CDR3*

Choose where to save (defaults to your Desktop), then click **Generate Insert**.

Step 4 — Read the result

The annotated document opens automatically in your browser, colour-coded exactly like the reference inserts (gold chains, green scaffold, red brackets and restriction sites). The status box shows the beta seam and flags anything to check. Two files are saved to your folder: the annotated document (`_annotated_insert.html`) and a plain-text sequence file (`_output.txt`).

To save the annotated document as a PDF: in the browser, File → Print → Save as PDF.

Gene-name notes

The app handles allele suffixes and slashes automatically, but for reference: you can enter gene names with or without the *01 suffix (Stitchr defaults to *01). For a specific non-01 allele, type it in full (e.g. TRBV7 -9*03). Names with a slash (e.g. TRAV29/DV5) are fine to type as-is.

If something goes wrong

Symptom	Fix
"Tkinter is not installed"	brew install python-tk, then relaunch
"Stitchr failed"	Run pip install stitchr and stitchrdl -s HUMAN
Seam check says CHECK	Open the document; verify the beta junction shows E before DLRNV. Re-check the CDR3/J.
Alpha or beta CDR3 not found	A gene or CDR3 was likely mistyped. Double-check the six values.

The command-line method (Stitchr by hand + `lentiviral_insert_job.py`) is still available and documented in [Lentiviral_Insert_Guide.pdf](#). The GUI automates that same pipeline.